

Juan Camilo Gonzalez Cabrera

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Website GitHub ORCID LinkedIn

Research Interests

Computational and numerical algebraic geometry; computer algebra; numerical analysis; formal verification of algebraic and numerical computation.

Education

New York University, Courant Institute of Mathematical Sciences Expected May 2027

M.S. in Mathematics

Fall 2026 coursework: Algebra I, Numerical Methods I, Computer Assisted Formalization of Math

Prior coursework: Numerical Methods II, Algebra II, Ordinary Differential Equations

Universidad de los Andes 2019–2024

B.S. in Mathematics (advisor: Mauricio Velasco)

B.S. in Systems and Computing Engineering (advisor: Nicolás Cardozo)

Relevant coursework: Commutative Algebra, Stochastic Optimization, Calculus of Variations and Optimal Control, Big Data, Recommender Systems

Software

HomotopyGetsReal.jl Code · Docs

- Julia package for homotopy continuation, focused on the numerical decomposition of real algebraic curves and surfaces into cell complexes and on certifying real solutions of polynomial systems.
- Reimplements the critical-point-based decomposition pipeline of `Bertini_real` as a single, dependency-light Julia package built natively on `HomotopyContinuation.jl`; adds an optional generic-projection mode and a diagnostic isosingular deflation subsystem, validated against Hauenstein and Wampler’s published examples (including the Whitney umbrella) and a coordinate-level comparison against `Bertini_real`’s output on the astroid curve.
- Registered in Julia’s General package registry; released under the MIT license.

Talks

When Homotopy Gets Real: Numerical Decomposition of Real Algebraic Curves and Surfaces in Julia

ALBATROSS 2026 (ALgeBraic meThods foR pOlynomial System Solving), Sorbonne Université, Paris, France, September 2026

Manuscripts in Preparation

- Software paper on `HomotopyGetsReal.jl`.
- Dynamics of the Hastings-Powell system (joint work).

Technical Skills

Programming: Julia, Python, Java

Math Software: OSCAR, Macaulay2, Sage, msolve

Tools: NumPy, SciPy, Pandas, scikit-learn, PyTorch, TensorFlow, PySpark, SQL, Git

Professional Experience

Developer, GESINP S.A.S. Dec 2023 – Aug 2025

- Designed quantitative models for lease contract valuation using discounted cash-flow methods and sensitivity analysis.
- Engineered numerical routines for amortization optimization and rate-adjusted pricing.

- Automated portfolio-level lease performance analytics under multiple economic scenarios.

Teaching Experience

Recitation Leader (Teaching Assistant), NYU Courant 2025–Present

- Led weekly problem-solving sessions for two undergraduate calculus sections (60+ students total).
- Reinforced rigorous proof-based reasoning and quantitative problem-solving techniques.

Teaching Assistant, Universidad de los Andes 2022

- Led complementary problem sessions for Differential Calculus and Integral Calculus.

Leadership Experience

Founding Member and Vice President, LeMMA 2026–Present

Student association for Master's students in Mathematics and Mathematical Sciences, NYU Courant

Student Representative, Mathematics Department 2023

Universidad de los Andes

- Elected representative for undergraduate mathematics students to the departmental student council.
- Participated in monthly meetings with faculty leadership and student representatives to discuss academic policies, curriculum development, and student initiatives.